

Protect Private Information with this ML Model

Here's a simple machine learning model to help detect the personally identifiable information of your customers and maintain its confidentiality.



Data is a great resource and we all know that we need to be very careful about ensuring its privacy and confidentiality, especially if we have access to a large amount of customer data as an organisation. To ensure that we are not letting any PII or personally identifiable information out of the organisation, one popular strategy is to extract and identify this information in each file and mask it either with cryptography or with random characters. Many other strategies can be followed to extract this information. If the given data is always in a specific format, methods like regular expressions can be used. If that is not the case, but we have enough data, we can build a machine learning model. Generative AI models like OpenAI can be used if we don't have enough data to train, but this is an expensive solution.

So let's learn how to build an ML model for detecting PII. This is basically information that can be used to identify individuals — it may be their name, phone number, government identity information, address, etc. One strategy is to build and deploy a model for each of these categories. We will follow a better and a simpler approach, and build a generic model for identifying and detecting any kind of PII.

Let's get started. I am using a Google Colab notebook as the IDE here. You can use your local Jupyter Notebook or Anaconda environment or even a Visual Studio Code - Jupyter Notebook extension for this experiment.

If you are using Google Colab notebooks like me, open the following link and click on 'new notebook' as shown in Figure 1.